Subject: Session 3.5
Year. Month. Date. ( )

Problem 29.1

$A = U \Sigma V^T$

$A^TA = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T$

$AV = U\Sigma$

$$A^TA = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$|A^TA - \lambda I| = (2-\lambda)(1-\lambda) - 1 = 0$$
$$= 2 - 3\lambda + \lambda^2 - 1$$
$$= \lambda^2 - 3\lambda + 1$$
$$\rightarrow \lambda = \frac{3 \pm \sqrt{9-4}}{2} = \frac{3}{2} \pm \frac{\sqrt{5}}{2}$$

$$\rightarrow \sigma_1 = \frac{1+\sqrt{5}}{2}, \ \sigma_2 = \frac{\sqrt{5}-1}{2}$$

Problem 29.2

$A^TA = \Sigma^2$

$V = I$

$u_i = \frac{1}{\sigma_i} w_i$